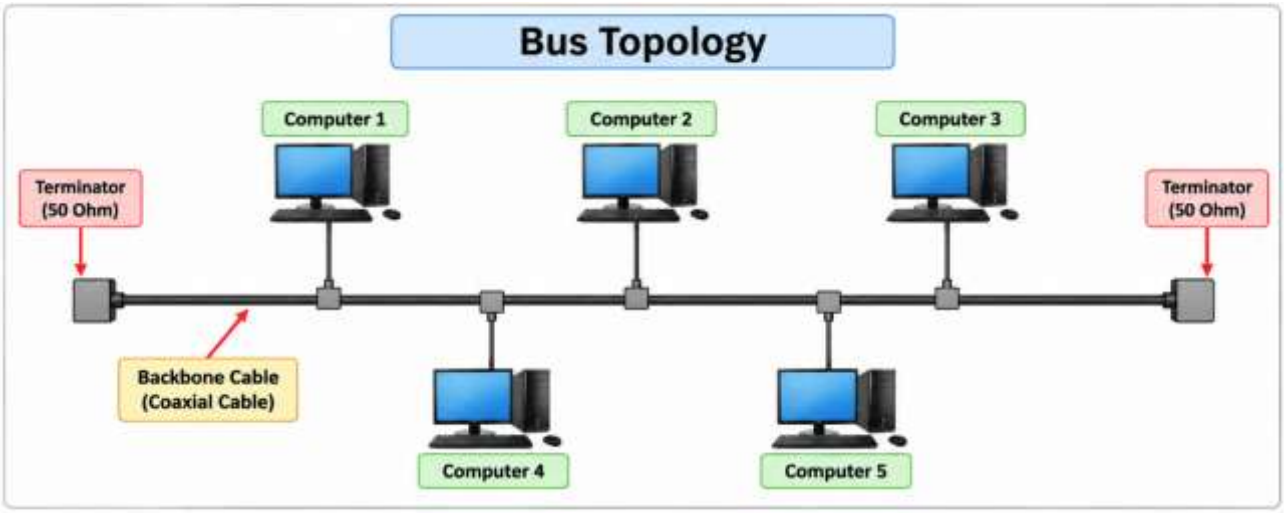
Network Topologies Theory

Q1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

Aim: To draw a bus network topology, label its key components, and identify one real-world application.

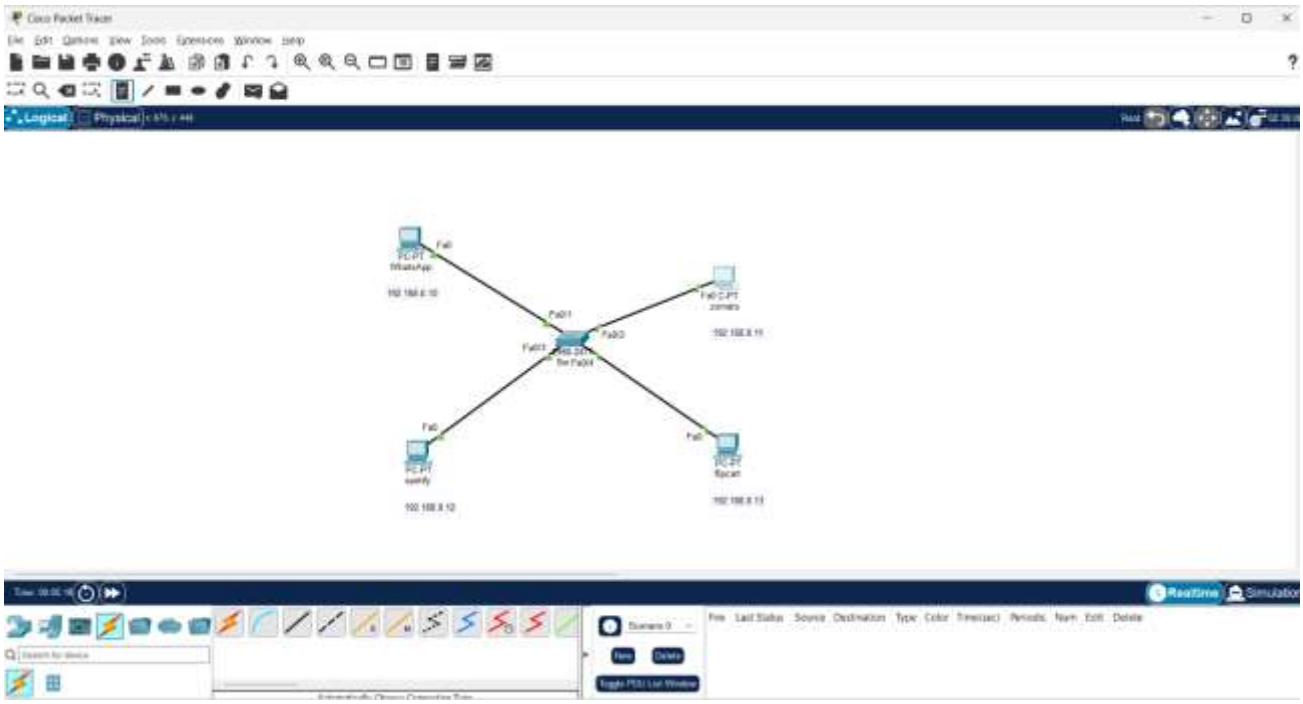
A bus topology is a network arrangement in which all devices are connected to a single main cable called the backbone cable. Data travels along this shared cable, and terminators at both ends help prevent signal reflections.



Key Components

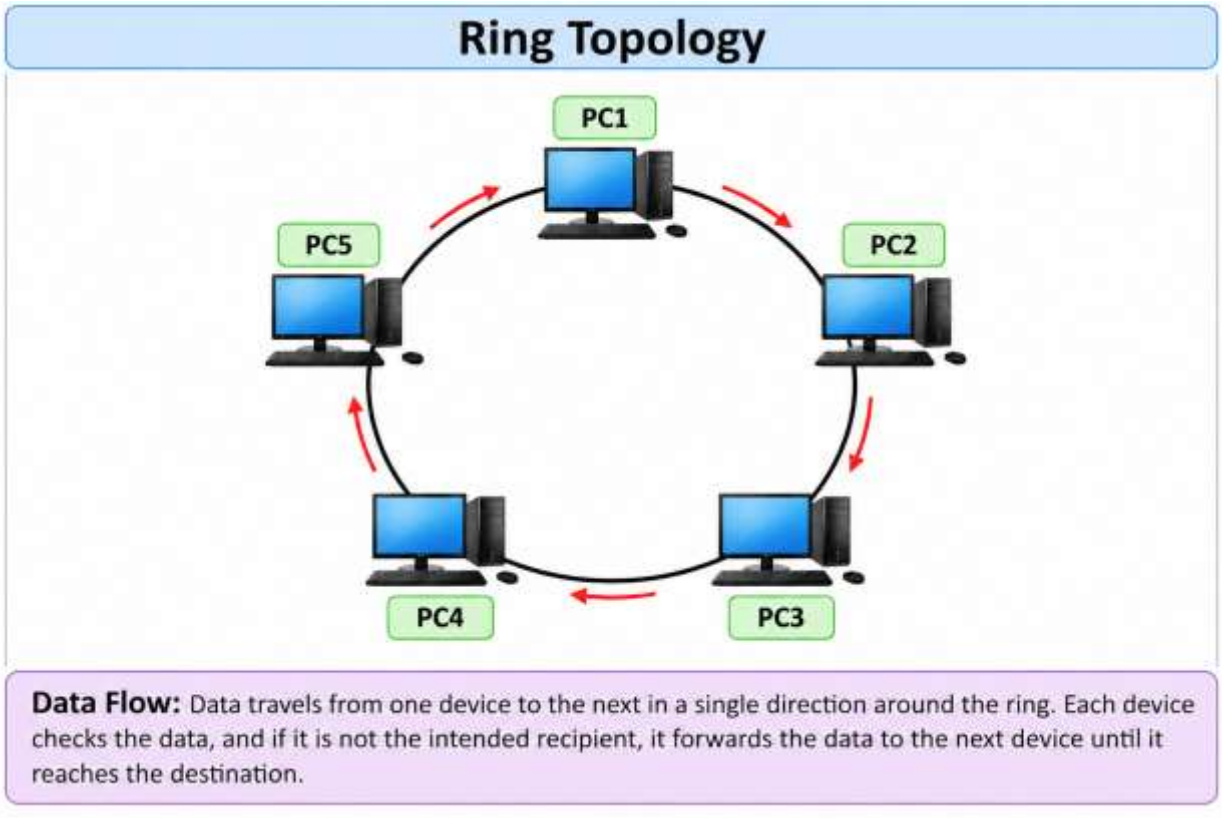
1. Backbone cable: The main coaxial cable connecting all computers.
2. Computers (nodes): Devices that communicate over the network.
3. T-connectors: Connect computers to the backbone cable.
4. Terminators: Installed at both ends of the cable to prevent signal reflections.

Q2 In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



**Q3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.**



Q4 Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format. Practical: Troubleshooting Slow College Wi-Fi

Topology	Advantages (2)	Disadvantages (2)
Mesh Topology	1. Highly reliable because multiple paths are available for data transmission. 2. Failure of one connection usually does not affect the entire network.	1. Expensive due to the large number of cables and connections. 2. Installation and maintenance are complex.
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one computer or its cable does not affect other devices.	1. Failure of the central switch can stop the entire network. 2. Requires more cabling than a bus topology.